



Neelam VidyaVihar, Sijoul, Mailam, Madhubani, Bihar – 847235

Website: <http://www.sandipuniversity.edu.in>

Email: info@sandipuniversity.edu.in

School of Computer Science & Engineering

Assignment: 03

Subject Name: Introduction to Machine Learning
Semester: IIInd

Subject code: AIML304
Branch: AIML

Short questions:

1. What is the purpose of a machine learning model?
2. Define stratified k-fold cross-validation with an example?
3. What is the time complexity of computing mean and variance?
4. Explain the concept of one-hot encoding in feature engineering?
5. What is the space complexity of a feature-engineered dataset?
6. Define a test set with an example?
7. What is the purpose of data quality in machine learning?
8. Explain the concept of leave-one-out cross-validation?
9. What is the time complexity of feature selection?
10. Define a validation set with an example?
11. Define the K-Nearest Neighbor algorithm?
12. What is the time complexity of training a linear regression model?
13. Explain the concept of a confusion matrix?
14. What is the purpose of the SSE metric in regression?
15. Define logistic regression with an example?

Long Questions:

1. Discuss the role of cross-validation in preventing overfitting.
2. Explain the implementation of K-Nearest Neighbor for classification with an example?
3. Discuss the time and space complexity of linear regression with examples?
4. Describe the role of evaluation metrics in supervised learning with examples?
5. Explain the implementation of logistic regression for binary classification?
6. Discuss the time and space complexity of Support Vector Machines?
7. Describe the implementation of a confusion matrix for classification evaluation?
8. Explain the importance of precision, recall, and F-Score in classification?
9. Discuss the applications of supervised learning in real-world scenarios?
10. Discuss the role of cross-validation in preventing overfitting?